Think before you compute

- Conservation laws and the numerical solution of the Navier–Stokes equations
- Transport equations
- *Brief intro to Basilisk coding framework*

First coding steps

- Implementing basic transport equations in Basilisk C.
- Using headers in Basilisk, modular code structure, problem setup, and compilation
- 1st Working Assignment

Next coding steps

- Solving Navier–Stokes equations in Basilisk C: 2nd Working Assignment

Interface Dynamics

- Interface tracking methods (VoF, level set, phase-field approaches) and numerical strategies
- Hands-on tutorial with interface-tracking to a simple two-phase problem: 3rd Working Assignment

Non-Newtonian flows

- Non-Newtonian flows: viscoelasticity.
- Coding exercises for viscoelastic fluids: 4th Working Assignment

Special topics

- Three-phase flows.
- Holey Sheets.

- Contact line dynamics.